

FIRST-YEAR CALCULUS, PART I
PRACTICE FOR THE FIRST MIDTERM

A substantial part of the first midterm will consist of the problems similar to the ones below. It is also recommended to revisit the lecture notes and work on the end-of-section problems in the textbook.

Exercise 1. Evaluate the following limits:

(a) $\lim_{x \rightarrow 2} \frac{x^2 - 6x + 8}{x^2 - 5x + 6}$

(h) $\lim_{x \rightarrow 0} \frac{2x - \sin x}{4x}$

(b) $\lim_{x \rightarrow 0} \frac{\sin(3x)}{x}$

(i) $\lim_{x \rightarrow +\infty} \frac{x^4 + 2x^2 + 2025}{x^5 - 1}$

(c) $\lim_{x \rightarrow +\infty} \frac{x^3 - 4x^2 + x + 1}{2x^3 + 5x^2 + 5x - 2}$

(j) $\lim_{x \rightarrow 6} \frac{x^2 - 13x + 42}{x^3 - 14x^2 + 48x}$

(d) $\lim_{x \rightarrow 7} \frac{x^2 - 15x + 56}{x^2 - 8x + 7}$

(k) $\lim_{x \rightarrow 5} \frac{\sin(x^2 - 25)}{x - 5}$

(e) $\lim_{x \rightarrow 0} \frac{\sin x}{3x}$

(l) $\lim_{x \rightarrow -\infty} \frac{2x^4 + x + 10}{-5x^4 + 10x^2 - 2x + 1}$

(f) $\lim_{x \rightarrow -\infty} \frac{x^5 + 3x^2 - x + 4}{x^3 + 10x + 12}$

(m) $\lim_{x \rightarrow 9} \frac{x^2 - 10x + 9}{x^3 - 17x^2 + 72x}$

(g) $\lim_{x \rightarrow 3} \frac{x^2 - 11x + 24}{x^3 - 3x^2}$

(n) $\lim_{x \rightarrow +\infty} x \sin\left(\frac{1}{x}\right)$

Exercise 2. In each item below, compute $f'(x)$ using the limit definition of derivative.

(a) $f(x) = \frac{x}{x+1}$

(f) $f(x) = \frac{1}{x^2 + 3x}$

(b) $f(x) = x^2 + 4x$

(g) $f(x) = \sin(2x)$

(c) $f(x) = \frac{1}{\sin(x)}$

(h) $f(x) = \sqrt{e^x}$

(d) $f(x) = \sqrt{2x+1}$

(i) $f(x) = e^{x^2}$

(e) $f(x) = e^{3x+5}$

(j) $f(x) = \ln(x^2)$

Exercise 3. In each item below, compute $f'(x)$ and determine the equation of the tangent line to the graph of f at $x = a$.

(a) $f(x) = (x^2 + 1)e^x$ and $a = 0$.

(f) $f(x) = x^3 \sin x$ and $a = 0$.

(b) $f(x) = \frac{\sin x}{x}$ and $a = \pi$.

(g) $f(x) = e^x \ln x$ and $a = 1$.

(c) $f(x) = x \ln x$ and $a = e$.

(h) $f(x) = \frac{\ln x}{x^2}$ and $a = 1$.

(d) $f(x) = 4x \cos x$ and $a = \pi/2$.

(i) $f(x) = \ln x \cos x$ and $a = e$.

(e) $f(x) = \sin x \cos x$ and $a = \pi$.

(j) $f(x) = \frac{\sin x}{x^{10}}$ and $a = 1$.